



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Features tracked

37

Missing Values

Review ratings imputed

Customer Data

- Age, Gender, Location
- Subscription Status
- Review Ratings

Purchase Details

- Item, Category, Amount
- Season, Size, Color
- Shipping Type

Behavior Metrics

- Discount Applied
- Purchase Frequency
- Previous Purchases

Data Preparation Process



Data Loading

Imported dataset using pandas



Initial Exploration

Checked structure and summary statistics



Data Cleaning

Handled missing values, standardized columns



Feature Engineering

Created age groups and purchase frequency



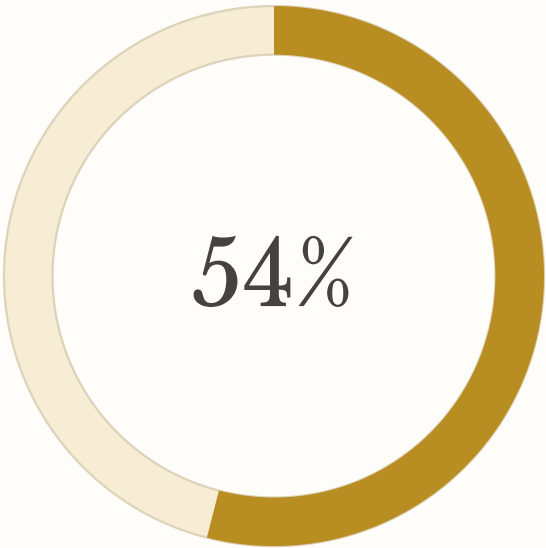
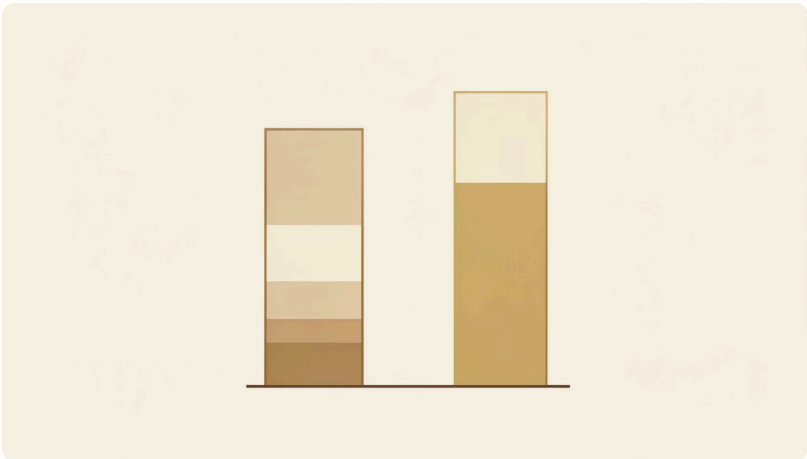
SQL Analysis: Revenue Insights

Revenue by Gender

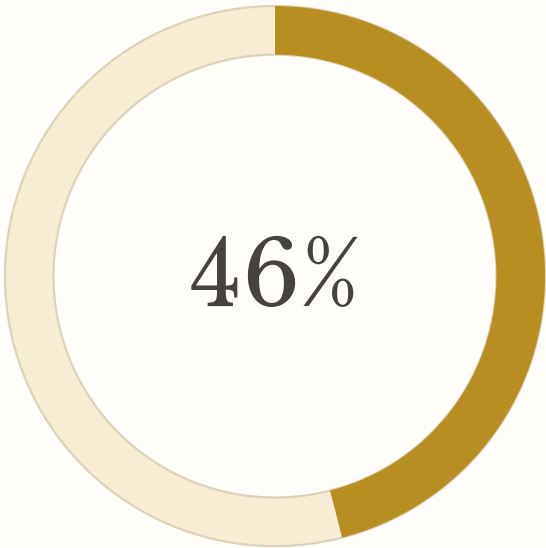
Female customers generated higher total revenue than male customers

Revenue by Age Group

Age groups analyzed for total revenue contribution patterns



Female Revenue Share



Male Revenue Share

Discount Strategy Analysis



High-Spending Discount Users

Customers who used discounts but spent above average purchase amount



Discount-Dependent Products

Top 5 products with highest percentage of discounted purchases



Product Performance

Top 5 Products by Rating

Products with highest average review ratings identified

Top 3 Products per Category

Most purchased products within each category



Shipping Type Comparison

Standard vs. Express shipping average purchase amounts compared

Repeat Buyers & Subscriptions

Customers with >5 purchases analyzed for subscription likelihood

Customer Segmentation

01

New Customers

First-time purchasers identified

02

Returning Customers

Multiple purchases, not yet loyal

03

Loyal Customers

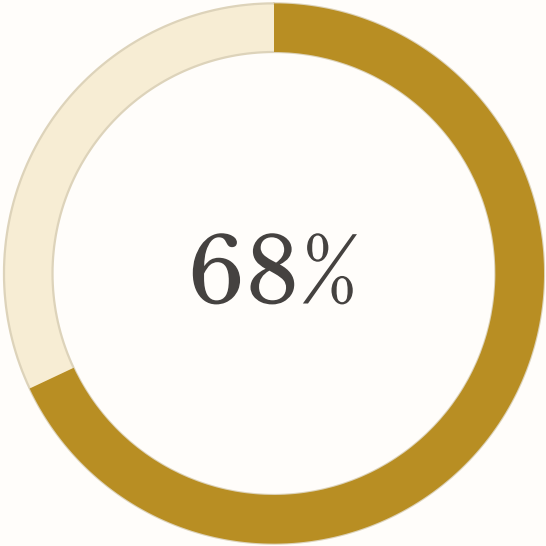
Consistent repeat buyers classified



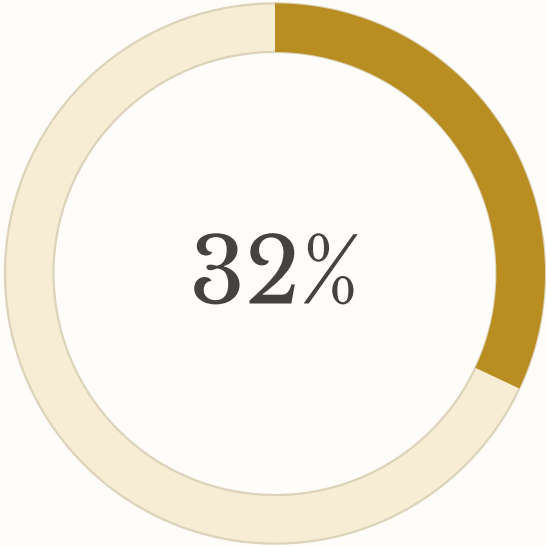
Subscription Analysis

Subscribers vs. Non-Subscribers

Average spend and total revenue compared across subscription status



Subscriber Revenue



Non-Subscriber Revenue



Power BI Dashboard

Interactive visualizations built to present insights across all analysis dimensions

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers



Customer Loyalty Programs

Reward repeat buyers to move into loyal segment



Review Discount Policy

Balance sales boosts with margin control



Product Positioning

Highlight top-rated and best-selling products in campaigns



Targeted Marketing

Focus on high-revenue age groups and express-shipping users